Shuai Liu

+1 (416) 820 2651 • liustark@hotmail.com • liushuai26.github.io

Machine Learning Engineer specializing in reinforcement learning, distributed training systems, and simulation-driven AI. Experienced in large-scale training infrastructure, multi-agent systems, and production ML deployment.

EXPERIENCE

Datatonic *Senior ML Engineer* *May 2026 – Present*

- Joined Datatonic as a Senior ML Engineer.

Hazel *Volunteer ML Engineer* *July 2025 – Feb 2026*

- Designed and implemented EnvLab, a **Gym-compatible RL training system** for the Hazel game engine, with a training-controlled step protocol ensuring deterministic synchronization between physics and agents.
- Built **parallel training fields** allowing N independent environments in a single physics world for **N× faster** sample collection, and integrated ONNX Runtime for in-engine inference deployment.

Freelancer *Independent ML Engineer* *Jan 2023 – Jan 2025*

- Designed and developed Beast Engine, an **AI-native** C++ 2D game engine with ECS architecture, Box2D physics, cross-platform Sokol rendering, ImGui visual editor, and native C++ hot-reload scripting.
- Developed dual Python integration paths for RL training: pybind11 native extension for **headless training** and TCP binary protocol for in-editor training, with snapshot/restore episode resets (<0.1ms).

bilibili *Machine Learning Engineer* *Jan 2021 – Dec 2022*

- Boosted data utilization efficiency by **1.5x** and **halved** training cycle time by developing a scalable, efficient **distributed training framework (up to 10,000 CPU cores and 80 GPUs)** with modular code structures and comprehensive documentation, deployed on bilibili's **internal cloud platform** to support diverse research needs.
- Leveraged optimized **PPO**, integrated **attention mechanism** into the network architecture, and engineered **heuristic rewards** to secure **1st place** in IJCAI2022 Neural MMO Challenge and top placements in multiple RL competitions—demonstrating the framework's effectiveness in single-agent, **multi-agent**, and self-play scenarios.
- Developed and integrated AI bots with **human-like behavior** in Unity games, enabling advanced gameplay with demonstrated human-level competencies.
- Collaborated with the live streaming team to design, develop, and successfully **launch live interactive games** for live streaming platform.

CreateAmind *Machine Learning Engineer* *Jul 2019 – Jan 2021*

- Optimized futures trading strategies for sub-5ms execution using **PPO** algorithms with **XGBoost**, achieving **expert-level performance** in real-world trading conditions.
- Enhanced football strategy modeling with Soft Q Network using entropy for exploration, and deployed distributed training via **Ray** on **Azure**, achieving **3× training efficiency** gains.
- Leveraged curriculum learning, self-play, and n-step solutions to achieve a **top 1% ranking** (11th of 1,138 teams) in Kaggle Google Research Football.

PROJECTS

BeaStrand, https://github.com/LiuShuai26/beastrand *2026*

- Built a **high-throughput distributed RL training framework** (3K LOC) with **comparable throughput to Sample Factory**.
- Engineered **async multi-process architecture** with shared-memory tensors and ZMQ IPC for zero-copy data transfer and batched GPU inference.
- Designed a **modular extension system** enabling custom algorithms (PPO-LSTM, AMP motion imitation) with zero changes to core code.

VSLAM Navigation *2025*

Developed a **ROS2** node on LIMO Robot using OpenCV ArUco, LiDAR, and wheel odometry to localize markers and autonomously navigate to their centroid.

Blasting Evaluation Demo, *Full-Stack ML System for Mining* *2024*

- Designed and deployed a **multi-stage ML pipeline** to evaluate blast performance from drone footage, enabling data-driven decision-making in mining operations.
- Applied **Mask R-CNN** for smoke plume **segmentation** and engineered features for blast quality prediction using **decision trees**.
- Developed the **full-stack** system: back-end model pipeline, data processing, and interactive Streamlit front-end for domain experts.

LastOrder-Dota2, *https://github.com/bilibili/LastOrder-Dota2* ***420 GitHub stars*** *2021*

The inference component of our Dota2 agent **outperforms 98%** of Dota players.

AWARDS

2026: **3rd Place** - Booster Soccer Showdown (34 teams)

2025: **1st Place** - Franka Golf Competition (59 teams)

2022: **1st Place** - IJCAI2022 Neural MMO Challenge (110 teams)

2022: **2nd Place** - CoG Football AI Competition (57 teams)

2022: **3rd Place** - IJCAI-ECAI AI Olympics (119 teams)

2020: **11th Place** - Kaggle Google Research Football (1,138 teams)

2017: CSC Scholarship - **State-financed** studying abroad

PUBLICATIONS

2025: **Shuai Liu**, Meng Cheng Lau. *Reduced-Order Model-Guided Reinforcement Learning for Demonstration-Free Humanoid Locomotion*. arXiv:2509.19023 (preprint).

2023: Yangkun Chen, Joseph Suarez, Junjie Zhang, **Shuai Liu**, et al. *Benchmarking Robustness and Generalization in Multi-Agent Systems: A Case Study on Neural MMO*. AAMAS 2023 (poster).

2020: Jingbin Liu, **Shuai Liu**, and Xinyang Gu. *Soft Q Network*. arXiv:1912.10891 (preprint).

2020: Jingbin Liu, Xinyang Gu, and **Shuai Liu**. *Policy Optimization Reinforcement Learning with Entropy Regularization*. arXiv:1912.01557 (preprint).

EDUCATION

Laurentian University **CA**
Master of Computational Science *2024–2025*

Oakland University **US**
Bachelor of Computer Science, Exchange Student *2017–2018*

Zhengzhou University of Light Industry **CN**
Bachelor of Computer Science *2015–2019*

SKILLS

Programming: Python (RL & ML development), C++ (engine & systems), C# (Unity), Go, SQL, Bash

ML & Reinforcement Learning: Multi-Agent RL, Self-Play, Curriculum Learning, Distributed RL, Model Deployment (ONNX Runtime)

Systems & Infrastructure: Ray, Horovod, Docker, Linux, AWS/Azure/GCP, ECS Architecture, ZeroMQ, CI/CD